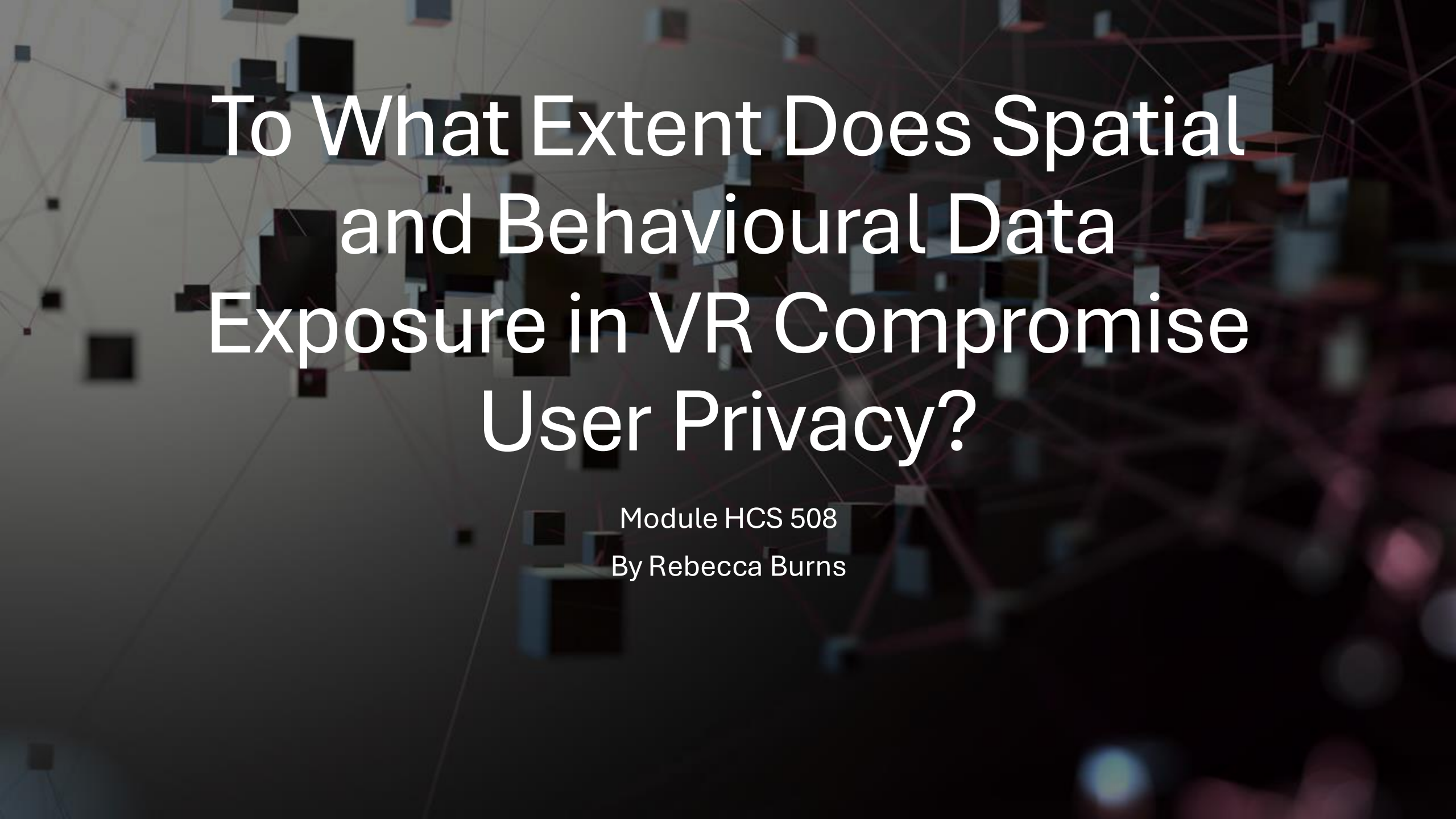




# To What Extent Does Spatial and Behavioural Data Exposure in VR Compromise User Privacy?

Module HCS 508

By Rebecca Burns

# Introduction

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This presentation explores how XR / VR spatial data technology works and the privacy risks it creates.



The MOTIVATION: VR is fully immersive, requiring continuous sensor, camera, and microphone data — capturing significantly more personal data than AR or MR, making it the greatest privacy risk



Understanding these privacy implications becomes essential for designing safer, more ethical immersive experiences.

# What is XR or Extended Reality?

**XR** Is the umbrella term for

**Augmented Reality or AR** - overlays digital elements onto the real world

**Mixed Reality or MR** - blends the two so virtual objects can interact with your physical environment

**Virtual Reality or VR**- fully replaces your surroundings with a digital world

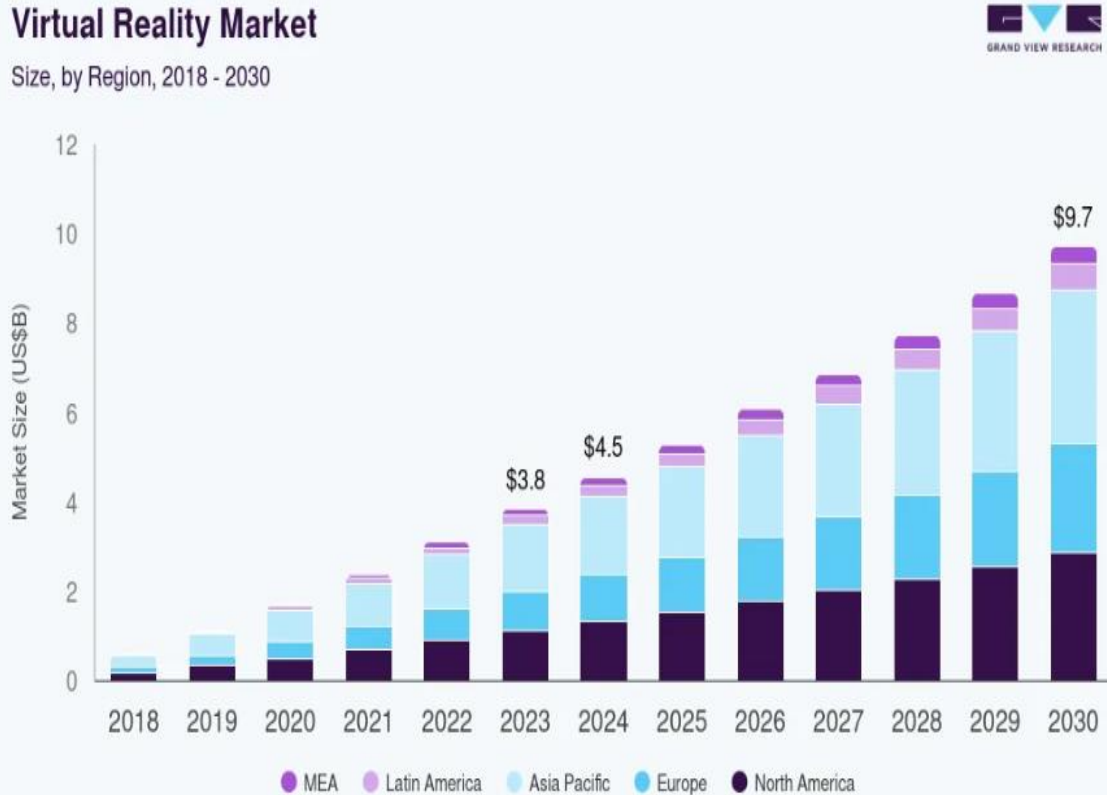


April, 2022

# Growing Trend of VR

## Virtual Reality Market

Size, by Region, 2018 - 2030



Market forecasts predict VR market to rise from \$60 billion in 2022 to over \$435 billion by 2030

The demand for VR is shifting from consumer entertainment to enterprise productivity

Grand View Research,  
2026

# What's Driving VR Growth?



- COVID-19 helped accelerate the CAGRs of AR and VR to 38.1 and 27.9% respectively.
- The growing integration of cloud computing and 5G
- VR training has been found to drive a 76% increase in learning effectiveness compared to traditional methods
- Increasing adoption from different industries



# 1. Healthcare

- Medical training and education
- **Simulated Procedure: Surgeons can practice complex operations in VR environments, honing their skills**
- Patient care and treatment aka (VRT). Numerous benefits:
  - Reduce pain perception
  - Lower anxiety



# Spatial Data in VR

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- Spatial data in VR is the 3D information a headset collects to understand your physical environment and your body movement within it



# Applied Feature: Room Boundary System

Type of Spatial Data

**Room boundaries**

Description

Dimensions and layout of your physical space

Real-World Use Case

Meta Quest's Guardian system to prevent collisions

The glowing blue lines outline the physical limits of the user's safe movement zone. The label "PLAY AREA" and warning triangle indicate where the headset's sensors have mapped the usable space.

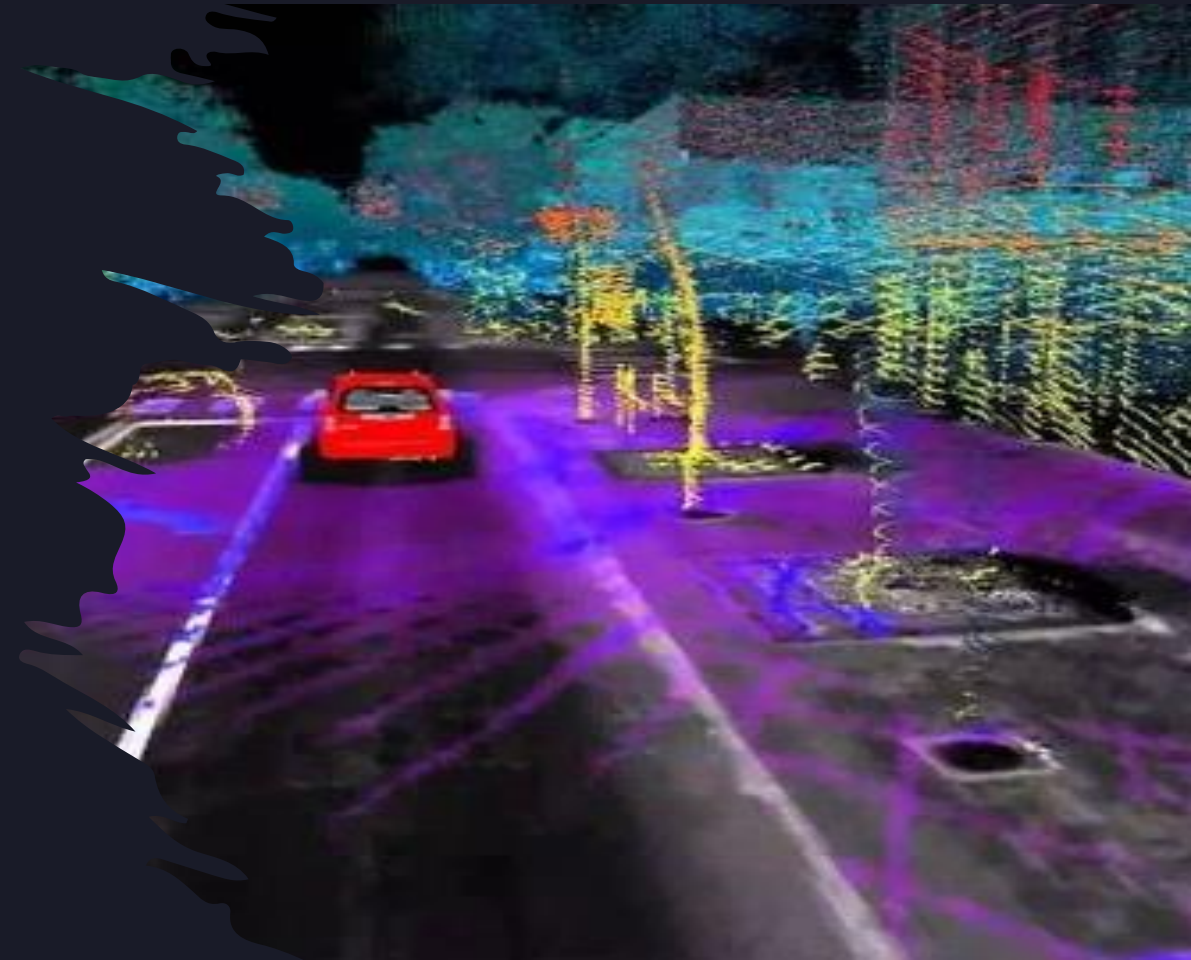




# SLAM

## Simultaneous Localization and Mapping

- Modern VR headsets (Head Mounted Devices, HMDs) depend on SLAM to function effectively, as it enables real time spatial mapping and positional tracking.
- Data captured through SLAM forms the basis for privacy exposure.



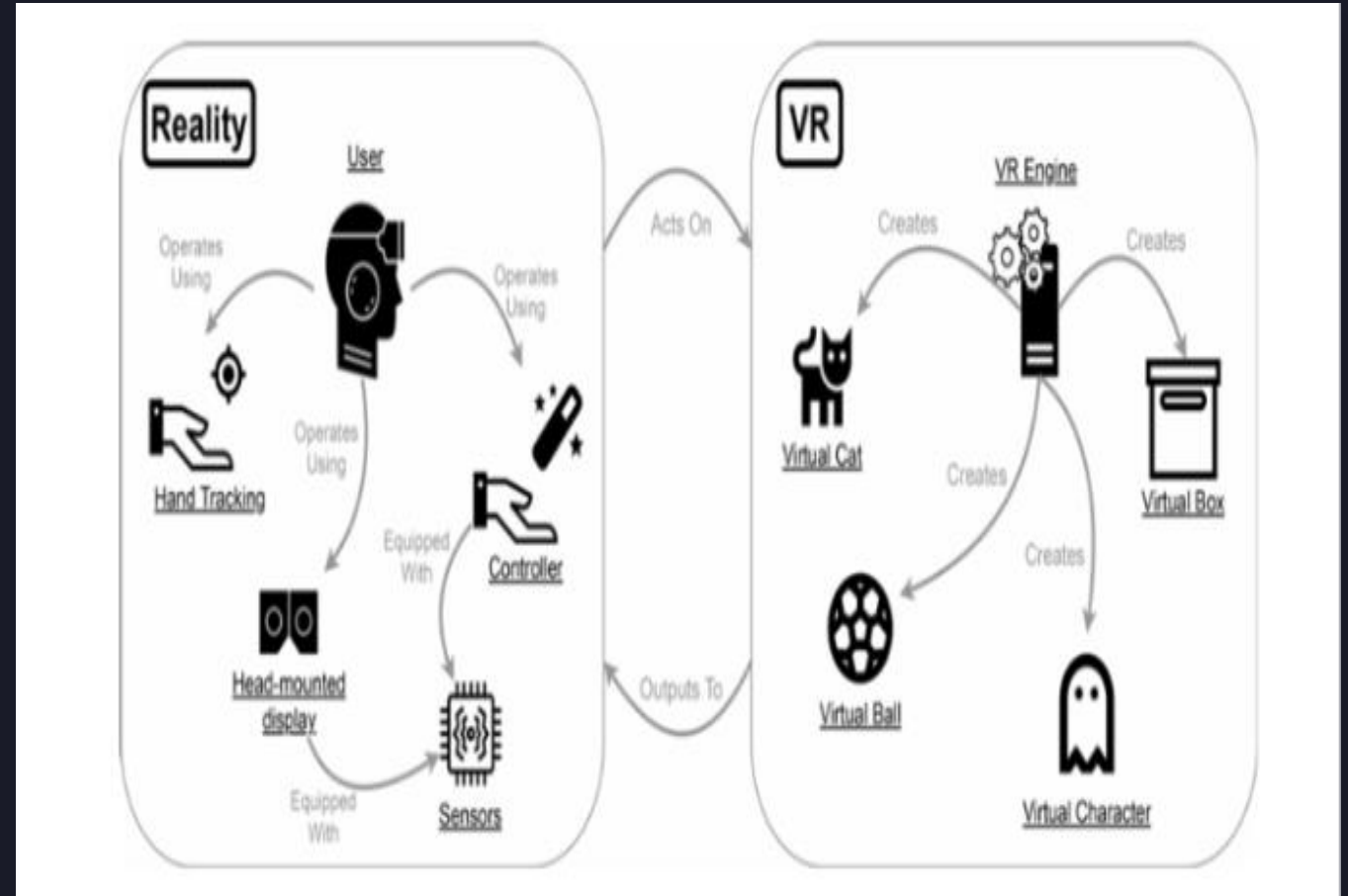
Flyability, 2024 SLAM example

# Spatial data vs SLAM

| Aspect     | Spatial Data                                                                      | SLAM (Simultaneous Localization and Mapping)                                               |
|------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Definition | Information describing the position, shape, and relationships of objects in space | A process that builds a map of an unknown environment while tracking the device's location |
| Nature     | Static or pre-recorded                                                            | Dynamic and real-time                                                                      |
| Purpose    | Represents spatial relationships and object positions                             | Enables navigation and mapping in real time                                                |

# VR paradigm diagram

- This diagram illustrates the continuous interaction loop between the user and the VR system. The user operates within reality using head-mounted displays, controllers, and sensors, while the VR engine creates and updates the virtual environment in response to those inputs.



Giaretta, 2024

**VR data loop: movement → capture → processing → feedback**

# Related Work and Spatial Data in XR

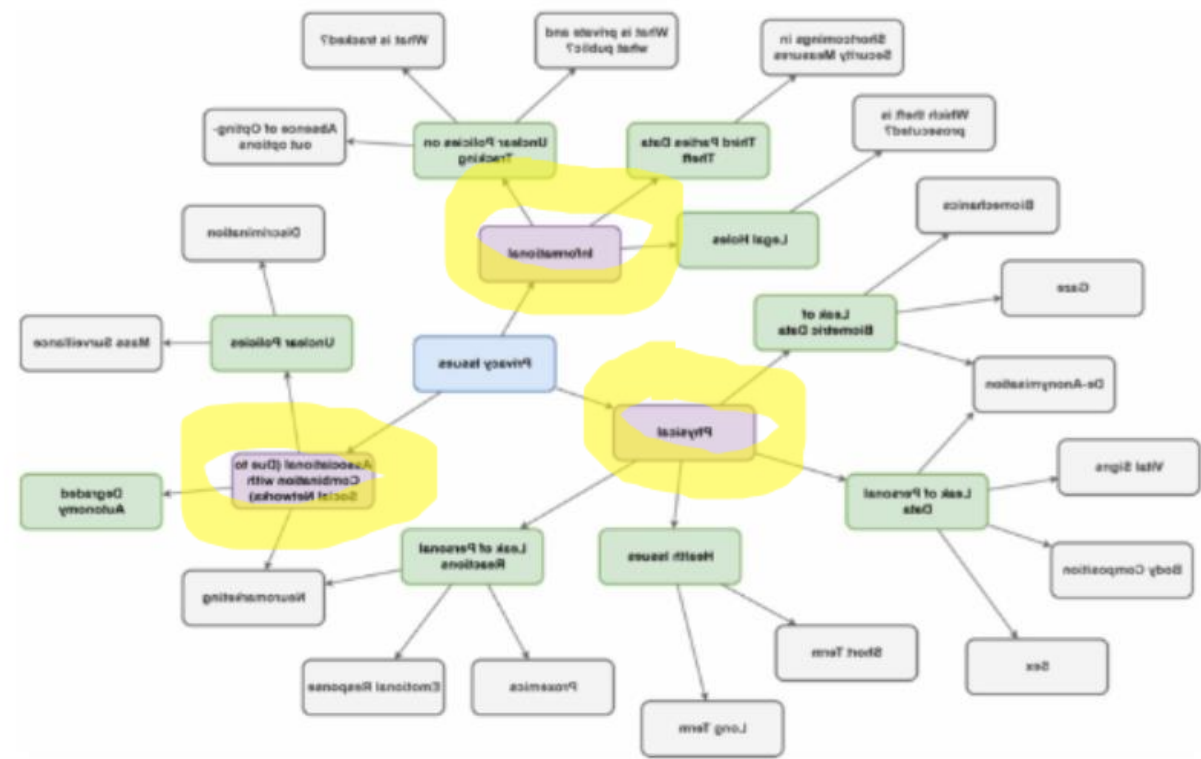
|                   |                                                                                                                                  |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Giaretta (2024)   | VR systems collect extensive spatial and behavioural data, increasing privacy risks compared to traditional digital technologies |
| Yang et al (2025) | SLAM generated 3D point clouds can reconstruct real world environments, exposing sensitive spatial information                   |
| Cai et al (2026)  | XR developers often underestimate the privacy risks associated with spatial data collection                                      |
| Hine et al (2024) | Existing regulatory frameworks struggle to fully address privacy risks in immersive XR systems                                   |

**Critical evaluation:** While these studies collectively highlight significant spatial privacy risks in XR, they identify threats in isolation without proposing integrated, empirically validated solutions. There is also limited consideration of user perspectives on acceptable data trade-offs, suggesting a gap between technical risk analysis and real-world user expectations.



# According to Giaretta (2024)

- This taxonomy provides a structured way to analyse privacy risks in VR systems;



According to Giaretta privacy risks can be arranged into 3 categories

- Focus on physical privacy

# Physical Privacy Immersive Systems

| Definition       | Data collected from the users physical body and movements during immersive interaction                |
|------------------|-------------------------------------------------------------------------------------------------------|
| Tracked elements | Head movement, hand position, posture locomotion, eye tracking, gesture patterns                      |
| Risks            | Exposure of biometric traits, health inferences, behavioural profiling, re-identification             |
| Example          | VR headsets analysing head and hand motion to maintain immersion and interaction                      |
| Ethical concern  | Users maybe unaware that functional tracking data can be repurposed into sensitive personal profiles. |



# Personal Data Leakage in VR

- SoK studies on VR privacy consistently identify spatial data as a major source of personal data leakage
  - Spatial data is classified as observed data – collected passively through sensors without users actively providing it, making it vulnerable to exploitation through third party sale or targeted advertising.
  - Munilla Garrido et al. (2023) classify spatial data as passively observed data in VR, showing that just minutes of telemetry can expose sensitive personal information about users and their environments
-

# Who Is Most Vulnerable?

## Children

- VR passively collects biometric data through sensors, cameras, and microphones — children cannot meaningfully understand or consent to this
- Parents significantly underestimate VR privacy risks — about half had never even considered them (Cao et al., 2024)
- Developing brains make it harder for children to distinguish virtual from real interactions, increasing susceptibility to manipulation

*As VR adoption grows across vulnerable groups and sensitive sectors, the urgency for privacy safeguards increases significantly.*



# Who Is Most Vulnerable?

## Sensitive Industries

- VR in **healthcare** exposes personal health information (PHI) through spatial and biometric data captured by sensors and cameras (Lake et al., 2024)
- Only **3 of 29 studies** in a scoping review addressed healthcare-specific cybersecurity risks in XR (JMIR, 2024)
- Information disclosure was identified as the **most critical threat** in clinical XR use — yet most mitigation strategies remain untested in real settings

*As VR adoption grows across vulnerable groups and sensitive sectors, the urgency for privacy safeguards increases significantly.*

# VR User Re-Identification Accuracy

83%

ACCURACY

Identified across multiple sessions of the **same activity**

80%

ACCURACY

Identified when performing a **different task**

78%

ACCURACY

Identified even with **deliberate obfuscation**

*These findings highlight the need for more robust technical measures to safeguard the behavioural privacy of VR users.*

**Critical Observation:** However, it is important to note that these accuracy rates were achieved under controlled laboratory conditions. Real-world environments introduce significantly more variability, which may affect the reliability and generalisability of these findings.

# Conclusion

So to what extent does spatial and behavioural data exposure in VR compromise user privacy ?- Significantly. The evidence shows.....

- SLAM enables VR immersion but continuously captures spatial and behavioural data that can reveal sensitive details about users and their environments.
- Re-identification studies show this data acts as a behavioural fingerprint – users can be identified with up to 83% accuracy even when trying to hide.
- Children and healthcare settings face the greatest exposure, yet privacy safeguards remain underdeveloped.

# Recommendations

- **Privacy-by-Design:** VR platforms should embed privacy safeguards into their architecture from the start, not bolt them on later. Cai et al. (2026) found that XR developers frequently underestimate privacy risks, reinforcing the need for built-in protections.
- **Data Minimisation:** Spatial and behavioural data collection should be limited to what is strictly necessary for functionality. Yang et al. (2025) demonstrated that SLAM-generated point clouds can reconstruct entire environments, while re-identification studies show users can be tracked with up to 83% accuracy — highlighting the dangers of excessive data collection.
- **Regulatory Alignment:** Existing regulatory frameworks are not fit for purpose when it comes to immersive technologies. Hine et al. (2024) found that current legislation struggles to address the unique privacy risks posed by XR, meaning frameworks like GDPR must be updated to explicitly cover spatial and environmental data.
- **Transparency and Consent:** Users must be clearly informed about what data is collected, how it is used, and whether it may be shared commercially. Munilla Garrido et al. (2023) classify spatial data as passively observed, meaning users may not even realise it is being captured, and Cao et al. (2024) found that parents significantly underestimate VR privacy risks for children.

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